

NATIONAL JUNIOR COLLEGE, SINGAPORE
Senior High 2
Preliminary Examination
Higher 2

CANDIDATE
NAME

BIOLOGY
CLASS

2BI2 / 2IPBi2

REGISTRATION
NUMBER

BIOLOGY

9648/01

Paper 1 Multiple Choice

19 Sep 2013

1 hour 15 minutes

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, Biology class, and registration number above and on the Answer Sheet provided.

There are forty questions on this paper. Answer all questions. For each question there are four possible answers A, B, C and D.

Choose the one you consider correct and record your choice in soft pencil on the separate Answer Sheet.

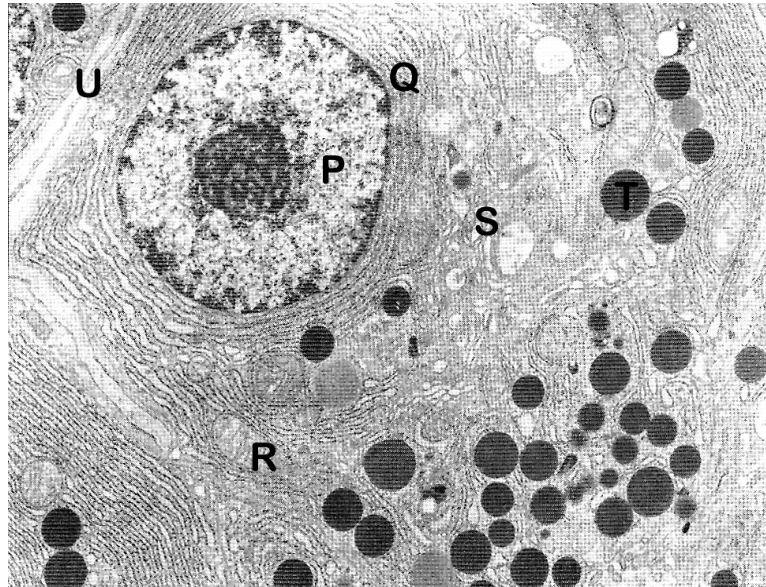
Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Calculators may be used.

- 1 The diagram shows an electron micrograph of a human cell.



Which of the following correctly identifies its structure-function adaptation?

	<i>function</i>	<i>structural adaptation</i>
A	enzyme production	P and U
B	antibody production	Q, S and T
C	phagocytosis	R
D	photosynthesis	T

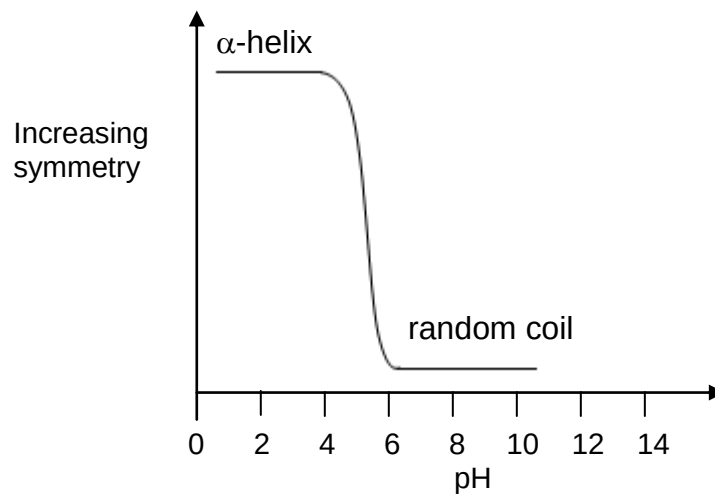
- 2 Most wild plants contain toxins that deter animals from eating them. A scientist discovered that a toxin produced by a certain plant was also toxic to the same plant if it was applied to the roots of the plant. As the first step on finding out why the plant was not normally killed by its own toxin, he fractionated some plant cells and found that the toxin was in the fraction that contained the largest cell organelle. He also found that the toxin was no longer toxic after it was heated.

Which of the following statements are consistent with the scientist's observations?

- I. The toxin was stored in the central vacuole.
- II. The toxin cannot cross the membrane of the organelle in which it is stored.
- III. The toxin was stored in chloroplast.
- IV. The toxin is likely to be lipid-soluble.
- V. The toxin may be an enzyme.

- A** I, II and V
- B** I, IV and V
- C** II, III and IV
- D** III, IV and V

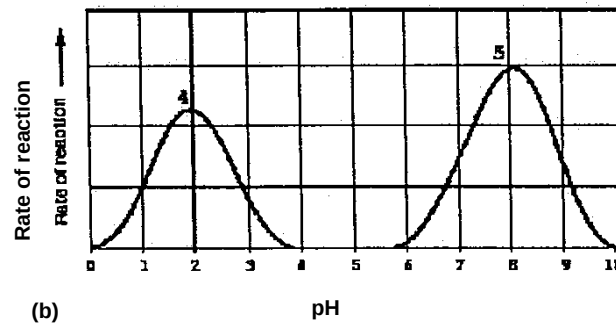
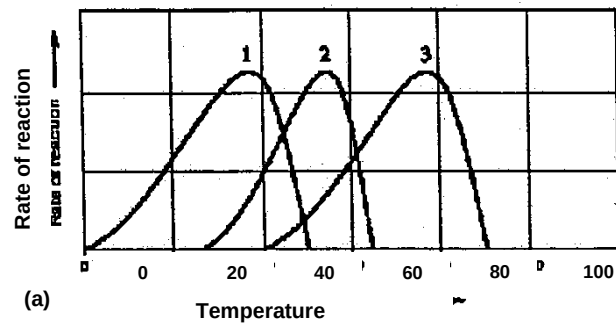
- 3 The graph shows the effect of pH on the structure of a protein which consists of entirely of repeating residues of one amino acid.



Which statement is true?

- A** At high acidity the protein loses its secondary structure.
- B** At high acidity the protein loses its tertiary structure.
- C** At high acidity the protein loses its tertiary structure.
- D** At low acidity the protein loses its secondary structure.

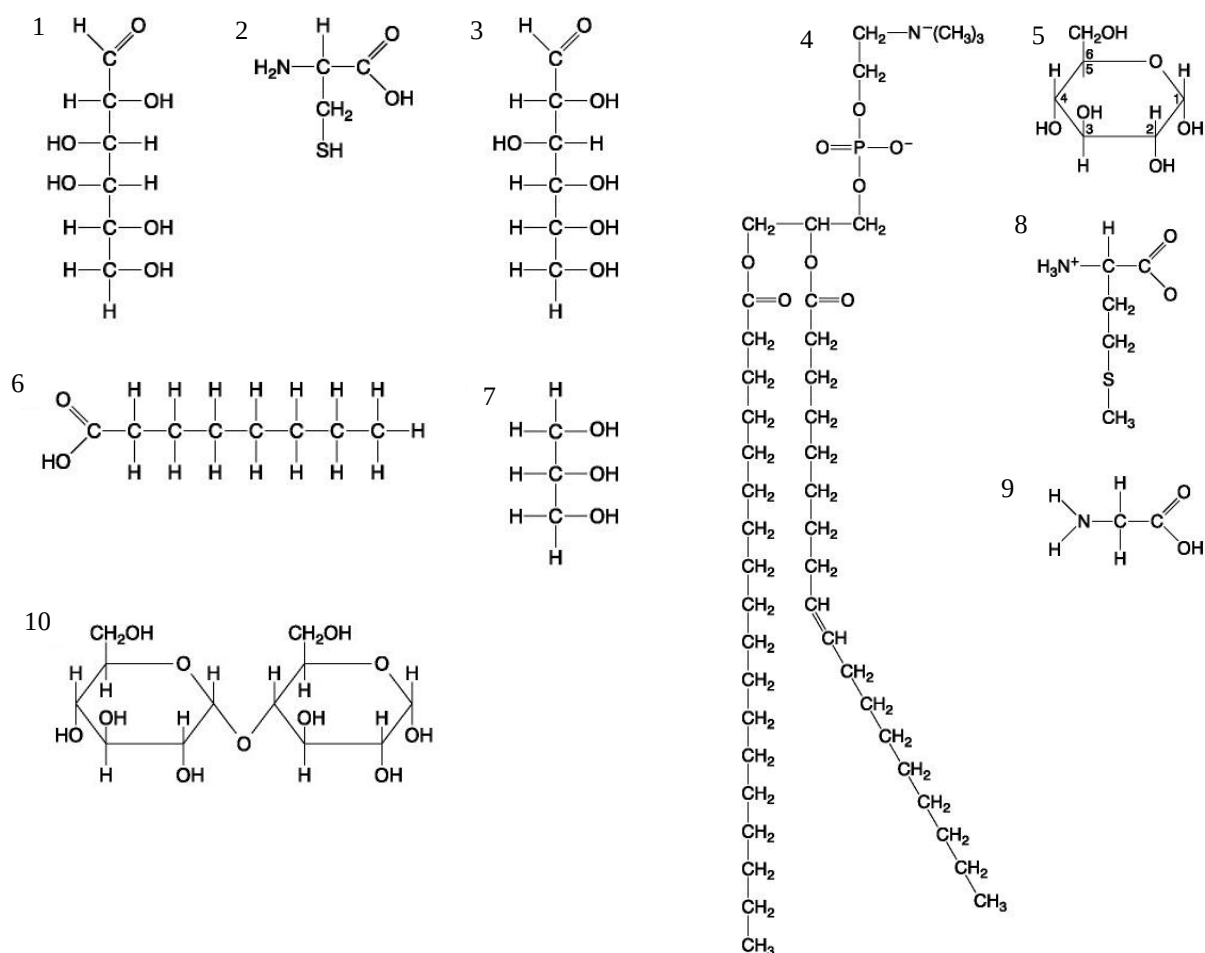
- 4 The following graphs show the activities of different enzymes (1-5) under different conditions:



Which statement is a correct explanation for the rate of reaction of different enzymes?

- A** Rate of reaction of enzyme 3 and its substrate is fastest at 58°C.
- B** At pH 2, most of the R groups at the active site of enzyme 4 are all negatively charged.
- C** At pH 8.1, substrate is bonded to the active site of enzyme 5 by hydrogen bonds only.
- D** At 60°C, several hydrogen bonds between R groups of enzyme 2 are broken.

5 The figure below shows some biomolecules numbered 1 to 10.



Which of the following statements concerning the biomolecules is/are **false**?

- I. Triacylglycerol may be made up of one molecule of 6 and three molecules of 7.
- II. Molecules 5 and 10 may be joined together by a glycosidic linkage to form a disaccharide.
- III. Molecules 1 and 3 are saturated fatty acids while molecule 4 has an unsaturated chain.
- IV. All of the molecules are monomers of biological macromolecules.

A	III only
B	I and II only
C	I, II and IV only
D	All of the above

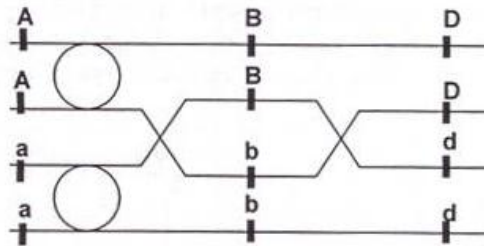
6 The following events occur during the mitotic cell cycle.

- I division of centromeres
- II proofreading mechanism in replication
- III duplication of chromosomes
- IV overlapping of non-kinetochore spindle fibres

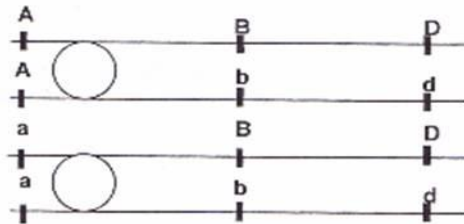
Which event(s) in mitosis directly ensure that the daughter cells are genetically identical to each other and to the parent cell?

- A I only
- B I and III
- C II and III
- D II and IV

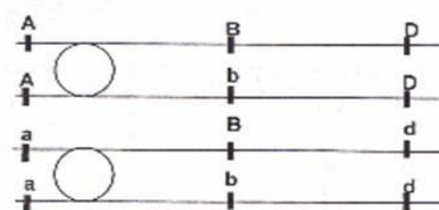
7 The diagram above shows a bivalent. Which of the following correctly represents the final products at the end of Meiosis II?



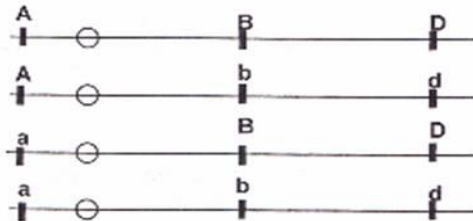
A



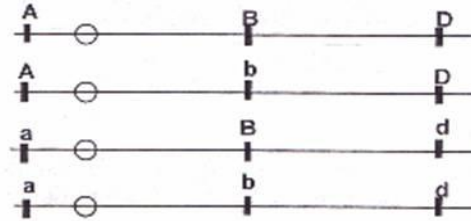
B



C



D



8 The mechanism of action of four drugs that inhibit DNA replication is stated below.

- Aphidicholine inhibits DNA polymerase
- Cytarabine is converted into a molecule that can substitute for a DNA nucleotide and also inhibits DNA repair mechanisms
- Epirubicin inhibits an enzyme involved in the unwinding of DNA and separation of strands
- Hydroxycarbamide inhibits an enzyme involved in the production of deoxyribonucleotides

Which row correctly matches a drug to an explanation of the mechanism of action?

	explanation of mechanism of action			
	decreased pool of available nucleotides inhibits chain elongation	DNA strands not available as templates for transcription	DNA damaged during replication and cell death occurs	exposed DNA template strands unable to be copied
A	aphidicholine	epirubicin	cytarabine	hydroxycarbamide
B	epirubicin	cytarabine	hydroxycarbamide	aphidicholine
C	hydroxycarbamide	aphidicholine	epirubicin	cytarabine
D	hydroxycarbamide	epirubicin	cytarabine	aphidicholine

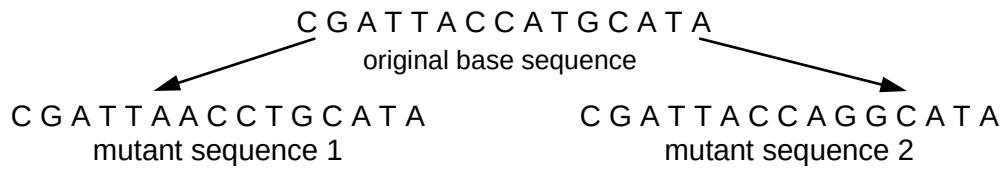
9 A polypeptide molecule contains the amino acid sequence, glycine – leucine – lysine – valine. The table shows the DNA codes for these amino acids.

glycine	leucine	lysine	valine
CCC	GAA	TTT	CAA

Which anticodons present on tRNA are needed for the synthesis of this polypeptide?

- A** CCC GAA TTT CAA
- B** CCC GAA UUU CAA
- C** GGG CUU AAA GUU
- D** GGG CUU UUU GUU

- 10 The diagram shows the results of two types of mutation.



What types of gene mutation produce mutant sequences 1 and 2?

1

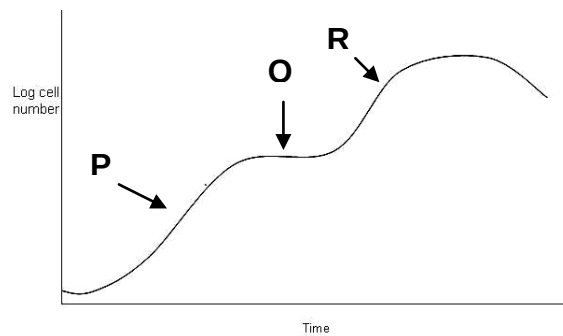
2

- | | | |
|----------|--------------|--------------|
| A | deletion | insertion |
| B | insertion | deletion |
| C | inversion | substitution |
| D | substitution | insertion |

- 11 Which of the following mutations will bring about active transcription of the lac operon in *E. coli*?

- A** A mutation in the promoter that decreases the affinity of the repressor.
- B** A mutation in the repressor gene that strengthens the affinity of the repressor for the inducer.
- C** A mutation in the repressor gene that weakens the affinity of the repressor for the operator.
- D** All of the above

- 12 *E. coli* bacteria are grown in a culture of nutrients, which includes glucose and lactose as the main source of carbon-based nutrient.



Which of the following statements account for the growth curve showing the reproduction of *E. coli* bacteria?

	P	Q	R
A	Lactose is metabolised	Production of enzymes for lactose metabolism is repressed	Glucose is metabolised
B	Lactose is metabolised	Enzymes for lactose metabolism is produced	Glucose is metabolized
C	Glucose is metabolised	Production of enzymes for lactose metabolism is repressed	Lactose is metabolised
D	Glucose is metabolised	Enzymes for lactose metabolism is produced	Lactose is metabolised

13 Which of the following statement about chromosome structure is true?

- I Linker DNA refers to the linear double-stranded DNA between adjacent nucleosomes.
- II The 30 nm chromatin fibre has a solenoid structure which forms due to attraction between histone tails and linker DNA.
- III The 300 nm chromatin fibre is attached to multiple locations on a central protein scaffold made of histone protein H1.
- IV Each nucleosome is made of linear DNA wrapping around four histone proteins.

- A** I and II
- B** I and IV
- C** II and III
- D** III and IV

14 Four different genes are regulated in different ways.

Gene 1 undergoes tissue-specific patterns of alternative splicing

Gene 2 is part of a group of structural genes controlled by the same regulatory sequences

Gene 3 is in some circumstances subject to methylation

Gene 4 codes for a repressor protein which acts at an operator site close by

Which combination correctly identifies the gene regulatory steps involving prokaryotes and eukaryotes?

	<i>prokaryotic</i>	<i>eukaryotic</i>
A	1 and 2	3 and 4
B	1 and 3	2 and 4
C	2 and 3	1 and 4
D	2 and 4	1 and 3

15 Which choice presents items in the proper order (earliest to latest) in which they would be used in the process of gene expression resulting in the formation of a protein?

- A** polyribosome, promoter, spliceosome, primary transcript
- B** spliceosome, promoter, primary transcript, polyribosome
- C** promoter, primary transcript, spliceosome, polyribosome
- D** primary transcript, spliceosome, promoter, polyribosome

- 16** The hepatitis C virus is a single-stranded, enveloped, positive sense RNA virus, which attack the liver cells of human beings.

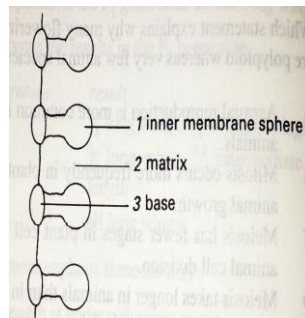
Which of the following correctly describe the reproductive cycle of hepatitis C virus?

- A** Fusion of the viral envelope with the host cell membrane, uncoating to release viral genome, followed by synthesis of complementary RNA for translation.
 - B** Fusion of the viral envelope with the host cell membrane, uncoating to release viral genome, followed by protein synthesis.
 - C** Receptor mediated endocytosis, uncoating to release viral genome, followed by reverse transcription.
 - D** Receptor mediated endocytosis, uncoating to release viral genome, followed by transcription.
- 17** In a hypothetical situation, DNA from T4 phage is mixed with proteins from a lambda phage, resulting in a hybrid virus. If this hybrid virus were to infect a bacterium and reproduce, what would the resulting "offspring" viruses be like?
- A** T4 phage
 - B** Lambda phage
 - C** A hybrid virus with T4 DNA and lambda protein
 - D** A hybrid virus with lambda DNA and T4 protein
- 18** BRCA1 is expressed in the cells of breast and other tissue, where it helps repair damaged DNA, or destroy cells if DNA cannot be repaired. BRCA1 gene is on chromosome 17 of the human genome.

Which individual is most likely to develop breast cancer?

- A** Has a frameshift mutation of the BRCA1 allele on each chromosome 17.
- B** Has a BRCA1 allele on each chromosome 17.
- C** Has a BRCA1 allele on one chromosome 17.
- D** Has a BRCA1 allele on one chromosome 17 and another, by translocation, on one chromosome 12.

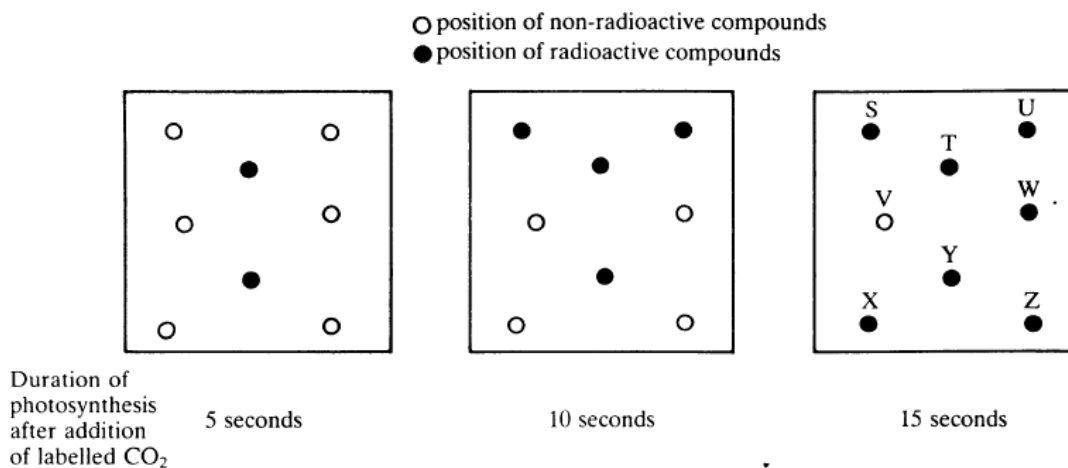
19 The diagram shows stalked particles on part of a crista membrane in a mitochondrion.



What occurs in each of the numbered regions?

	1	2	3
A	ADP synthesis	electron transport	Krebs cycle
B	ATP synthesis	glycolysis	Krebs cycle
C	ATP synthesis	Krebs cycle	electron transport
D	ATP synthesis	Krebs cycle	glycolysis

20 The diagram represents autoradiograph two-dimensional chromatograms of different algal extracts. The extracts were obtained after algal suspensions, which had been exposed to radioactive CO_2 for different length of time in conditions suitable for photosynthesis, were plunged into hot ethanol to prevent further biochemical reactions from occurring.



Assuming that all techniques and conditions (apart from time) were standardised, which of the following conclusions is justified from these data?

- A compound X must be sucrose
- B compound V must be an intermediate in the photosynthetic pathway
- C compound T precedes compound Y in the photosynthetic pathway
- D compound U precedes compound X in the photosynthetic pathway

- 21** In tomato plants, 'purple pigment' and 'hairy stem' are genetically determined and the genes are tightly linked. Pure breeding, purple, hairy plants crossed with pure breeding green, hairless plants produced only purple, hairy plants in the F_1 generation.

These F_1 plants were then crossed to produce the F_2 generation and a total of 500 F_2 generation plants were produced. Which of the following best describes the number of offsprings obtained from this cross?

	<i>purple, hairy</i>	<i>purple, hairless</i>	<i>green, hairy</i>	<i>green, hairless</i>
A	187	48	52	213
B	204	205	45	46
C	282	218	0	0
D	467	0	0	33

- 22** In a series of mapping experiments, the recombination frequencies for four different linked genes of *Drosophila* were determined as shown below.

<i>b</i>	0			
<i>cn</i>	9	0		
<i>rb</i>	3.5	6.5	0	
<i>vg</i>	19	9.0	16	0
	<i>b</i>	<i>cn</i>	<i>rb</i>	<i>vg</i>

b = black body
cn = cinnabar eyes
rb = reduced bristles
vg = vestigial wings

The numbers in the boxes are the recombination frequencies in between the genes (in percent).

What is the order of these genes on a chromosome map?

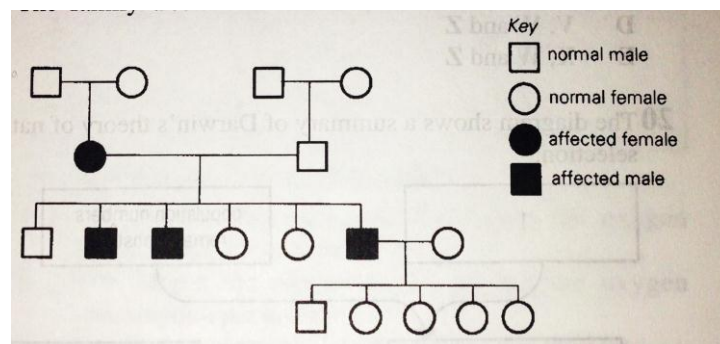
- A** *b-rb-cn-vg*
B *cn-rb-b-vg*
C *rb-cn-vg-b*
D *vg-cn-b-rb*

- 23** The feather colors in budgies (a kind of bird) are determined by two pairs of unlinked genes Y/y and allele B/b. The genotype of green color feather is Y_B_, blue yyB_, and white __bb (where _ indicates the presence of either the dominant or recessive allele).

If two heterozygous budgies are mated, what proportion of phenotypes can be expected in the progeny?

- A** 9 green color feathers : 7 white color feathers
- B** 9 green color feathers : 3 blue feathers : 4 white feathers
- C** 15 green color feathers : 1 blue feathers
- D** 15 green color feathers : 1 white feathers

- 24** The family tree shows the inheritance of a human defect.



What is the most likely genetic basis for the occurrence of this defect?

- A** autosomal dominant
- B** autosomal recessive
- C** sex-linked dominant
- D** sex-linked recessive

- 25 Test crossing of some tomato plants gave the following results.

plants with round leaves and are tall	157
plants with round leaves and are dwarf	143
plants with oval leaves and are tall	140
plants with oval leaves and are dwarf	160

The expected phenotypic ratio from such test cross is 1 : 1 : 1 : 1. A χ^2 (chi-squared) test was performed to test the significance of the difference between the observed and expected results. The calculated $\chi^2 = 8.36$.

distribution of χ^2			
number of degrees of freedom (ν)	probability		
	0.1	0.05	0.01
1	2.71	3.84	6.64
2	4.60	5.99	9.21
3	6.25	7.82	11.34
4	7.78	9.49	13.28

Which combination correctly describes the result of the χ^2 test?

	number of degrees of freedom (ν)	probability	significance of difference between observed and expected results
A	3	< 0.05	not significant
B	3	< 0.05	significant
C	4	> 0.05	not significant
D	4	> 0.05	significant

26 The followings are possible events that may lead to speciation.

- I Gene mutation
- II Increased gene flow
- III Natural selection
- IV Geographical isolation
- V Habitat differentiation within the same geographical location

The correct order of events that may lead to sympatric speciation is

- A IV, I, III
- B IV, II, III
- C V, I, III
- D V, II, III

27 Members of two different species possess a similar-looking structure that they use in a similar fashion to perform the same function. Which information would best help distinguish between an explanation based on homology versus one based on convergent evolution?

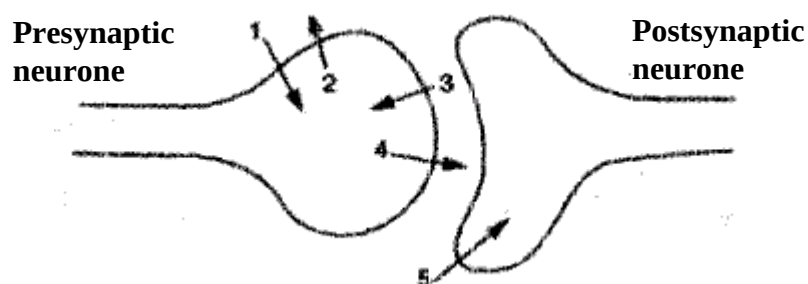
- A Both species are well adapted to their particular environments.
- B The sizes of the structures in adult members of both species are similar.
- C The two species live at great distance from each other.
- D The two species share many proteins in common, and the nucleotide sequences that code for these proteins are almost identical.

28 Which of the following would provide the best information for distinguishing phylogenetic relationships between several species that are almost identical in anatomy?

- A Comparative embryology.
- B Homologous structures.
- C Molecular comparisons of DNA and amino acid sequence.
- D The fossil records.

- 29** The diagram shows the sequence of events occurring as an action potential arrives at a synapse.

The numbered arrows represent the movement of substances across the membranes.



What are the substances moving across the membranes?

	1	2	3	4	5
A	K ⁺	Na ⁺	acetylcholine	Ca ²⁺	K ⁺
B	K ⁺	Na ⁺	K ⁺	Ca ²⁺	acetylcholine
C	Na ⁺	K ⁺	Ca ²⁺	acetylcholine	Na ⁺
D	Na ⁺	K ⁺	Na ⁺	acetylcholine	Ca ²⁺

- 30** The post-synaptic membrane of a nerve may be stimulated by certain neurotransmitters to permit the influx of chloride ions into the cell. This process will result in

- A** an action potential.
- B** membrane depolarization.
- C** the production of an excitatory post-synaptic potential.
- D** the production of an inhibitory post-synaptic potential.

- 31** Two organs secrete substances which affect the body.

organ 1 —————> product 1

organ 2 —————> product 2

Negative feedback control of product 2 would be achieved if

- A** product 1 counteracts product 2.
- B** product 2 inhibits organ 1 and product 1 inhibits organ 2.
- C** product 2 inhibits organ 1 and product 1 stimulates organ 2.
- D** product 2 stimulates organ 1 and product 1 stimulates organ 2.

32 Which of the following is a similarity between G protein coupled receptors and receptors tyrosine kinase?

- A** Conformational change that results from binding of ligand molecule.
- B** Formation of a dimer following binding of a signal molecule.
- C** Ligand-binding sites are specific for steroid hormones.
- D** Seven regions of the receptor span the plasma membrane.

33 Which feature of stem cells obtained from blood in the umbilical cord enables their use in the treatment of a variety of blood disorders?

- A** They are totipotent.
- B** They can differentiate into any type of blood cell.
- C** They can differentiate into stromal stem cells cells.
- D** They can renew indefinitely to replace defective blood cells.

34 One therapy to treat β -thalassaemia is to transplant bone marrow cells from a genetically compatible donor into a patient. A potential gene therapy involves adding the normal, dominant allele for β -globin to the patient's cells.

Which of the following would ensure that the normal allele is passed on to the next generation?

- A** Using a liposome to introduce the normal β -globin allele into bone marrow cells.
- B** Using a retrovirus to introduce the normal β -globin allele into a germ cell.
- C** Using an adeno-associated virus to introduce the normal β -globin allele into bone marrow cells.
- D** Using an adenovirus to introduce the normal β -globin allele into a germ cell.

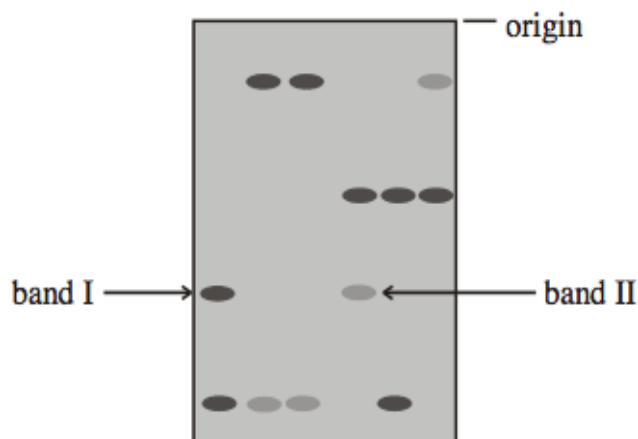
35 Each of the statements is concerned with the success or failure of gene therapy as a permanent cure for genetic disorders.

- I** Somatic cells cannot pass the modified gene on to any offspring
- II** The treatment does not last long as the treated cells die and are replaced
- III** The transfection efficiency of non-viral vectors is rather low.

Which statements explain the limitation of gene therapy as a permanent cure for genetic disorders in human population?

- A** I and II
- B** I and III
- C** II and III
- D** All of the above

- 36 The diagram below shows the results of DNA profiling using gel electrophoresis.



What conclusion can be drawn about the DNA in bands I and II?

- A The DNA in the two bands has the same base sequence.
- B The DNA in the two bands consists of fragments of the same length.
- C The DNA in the two bands has the same ratio of bases.
- D The DNA in the two bands came from the same source.

Refer to the following information to answer **Q37** and **38**.

Erythropoietin (EPO) is a small signaling molecule produced by certain kidney cells, which stimulates the production and differentiation of red blood cells. A scientist is interested to clone the cDNA of EPO receptor (EPO-R) from mouse and express it in suitable human host cells. The first step in library construction is cDNA synthesis.

- 37 Which of the following components show the correct combinations that are required for synthesizing a collection of cDNAs containing copies of EPO-R cDNA?

	<i>DNA ligase</i>	<i>DNA polymerase</i>	<i>RNA polymerase</i>	<i>reverse transcriptase</i>
A	✓	✗	✓	✗
B	✓	✓	✗	✗
C	✗	✗	✓	✓
D	✗	✓	✗	✓

- 38** The cDNAs synthesized in this mixture were converted into double-stranded DNA molecules supplied with Xho restriction endonuclease specific cohesive ends and ligated into the Xho site of a mammalian expression plasmid.

Which of the following sequence should be present in this plasmid if it is to express inserted cDNAs in human cells?

- A** A bacterial promoter downstream of the Xho site
- B** A bacterial promoter upstream of the Xho site.
- C** A mammalian promoter downstream of the Xho site.
- D** A mammalian promoter upstream of the Xho site.

- 39** Which of the following is **not** an example of genetically modified organisms?

- A** Bt corn plants that are insect-resistant.
- B** Delayed ripening in tomato using anti-sense technology.
- C** Golden Rice that produce high levels of beta-carotene.
- D** Inbred Salmon that grow to adult size quickly.

- 40** Plants are more readily manipulated by genetic engineering than are animal cells because

- A** a somatic plant cell can often give rise to a complete plant.
- B** more vectors are available for transferring recombinant DNA into plant cells.
- C** plant cells have cellulose cell wall which allows substances to pass through easily.
- D** plant genes do not contain introns.

--- End of Paper ---

